

MERN STACK



MERN is a popular JavaScript stack used for building modern web applications. MERN gets its name from 4 technologies that make up the stack which are MongoDB, Express, React, Node.

Advanced Js

How Javascript Works

- hoisting
- strict mode
- closure
- synchronous and asynchronous programming

Event propagation

- bubbling
- capturing

Event loop

function curring

ajax and json

calling api using ajax

callback hell

promise

async await

fetch

modules

version

Typescript

- Why Typescript
- Basic Types
- Class and Interfaces

React Js

INTRODUCTION TO REACT JS

- What is React JS?
- What is SPA?
- DOM vs Virtual DOM
- Advantages and Disadvantages
- Key Features

ENVIRONMENTAL SETUP

- Node | NPM
- Installation of CLI
- Setup Project
- Directory Structure
- Code Editors
- How React JS Application Boot

BASIC FEATURES OF REACT JS

- React Concepts
- JSX and TSX
- Render Elements
- Function and Class Components
- Props and State
- Handling Events
- Dynamic Data Rendering
- Property Binding

KEY FEATURES OF REACT JS

- Conditional Rendering
- List and Keys
- Forms Handling
- Forms Validations

COMPONENT LIFECYCLE HOOK

- Understanding component lifecycle
- All Lifecycle Hooks

EVENT HANDLING REACT

- Understanding React Event System
- Passing arguments to event Handlers

NETWORK CALL

- Fetch
- Axios

CUSTOM SERVICES

- Introduction to Services
- Building a Service

LOCAL DATA STORAGE

- Local Storage
- Session Storage
- Cookies

ROUTING WITH REACT ROUTER

- Setting up React Router
- Configuring route with Route Component
- Making routes dynamic with Route Params
- Working with nested routes

- Link and NavLink
- Redirect Routes

UI COMPONENTS

- Material Design
- PrimeNG

INTRODUCTION TO REDUX

- Why Redux
- Install and setup
- Store, Reducer, actions
- Dispatcher
- High order Components
- mapStateToProps and map DispatchToProps usage

ADVANCE REDUX

- Async Actions
- Middleware
- Redux Thunk and Redux Saga

React Hooks

- Why We Need Hooks.
- Different Types Of Hooks
- Using State And Effect Hooks
- Usereducer, Userref Etc.
- Custom Hooks
- Rules Of Hooks

Third Party Modules

- Social Login
- Pagination
- Search
- Filter
- JWT Token
- File Upload
- Many More

Rest Js Testing

- Jest with Enzyme

Develop a CRUD Application in React Js

Node Js

Introduction to Node.js

- What is node.js
- A word on event loop
- Event driven architecture
- Blocking vs non-blocking code
- Node Process Model

- Setup node.js
- Node REPL
- Debugging node.js app

Modules in JS

- Understanding exports and require
- Creating modules
- Importing modules
- Quick tour on NPM
- Installing 3rd party modules

Events & streams

- Significance of Events
- Event Emitter class
- Emitting and listening to events
- Types of streams
- Working with streams
- Composing streams using pipe

Working with Process and File System

- Process object
- Handling exceptions at process level
- Listening to process events
- Working with file system using fs module
- Creating, Copying, Deleting files
- Creating, Copying, Deleting directories
- Watching for file changes

HTTP

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers

Express Js

Expressjs Introduction & Setup

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers
- Serving static files with express
- Express application generator

Routing

- Route Methods
- Route Paths
- Parameterized Routes
- Route Handlers
- Express Router

Middleware

- Types of middleware

1. Application
2. Router
3. Error
4. Built-in
5. Third-party
 - Error middleware
 - Body parser, cors, cookie parser, session management

Template Engines

- What are template engines
- EJS
- Jade
- Handlebars

Expressjs security

1. Authentication
 2. JWT
 3. Securing routes
- Debugging in Express
 - Realtime programming with Socket.io

Mongo DB

Introduction to MongoDB

- Installing MongoDB
- The current SQL/NoSQL landscape
- Document-oriented vs. other types of storage
- Mongo's featureset
- Common use-cases
- MongoDB databases
- MongoDB Collections
- MongoDB Documents

CRUD Operations in MongoDB

- Creating documents
 1. insert()
 2. update()
 3. save()
- Querying documents
 4. find()
- 5. Working with equality
- 6. Query operators
- 7. Building complex queries
 - Updating documents
 - Deleting documents
 - Limit
 - Order By

Introduction to Mongoose

- Word on ORM/ODM
- Installing mongoose
- Connecting to MongoDB from mongoose

Core concepts of Mongoose

- Understanding mongoose schemas and datatypes
- Working with Models
- Using modifiers in schema
- Using virtual fields
- Optimizing query performance by enabling indexes

Mern Application Deployment

- Build Application
- Deploy in Heroku



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